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Task-First Robotics: A Ground-Up Strategy Toward Autonomous Energy Facilities

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Abstract

This study is to systematically evaluate the feasibility of complementing oil and gas field operators' routine tasks with robotic systems, particularly within oil and gas plants and similar facility environments. Operators in these settings perform essential inspection rounds, detect anomalies using sensory input, and help maintain operational integrity. As robotics, artificial intelligence (AI), and Industrial Internet of Things (IIoT) technologies advance, the energy sector faces a growing need to understand which tasks can be automated and how. This study addresses a key gap in the literature by using a task-first, operator-centric methodology and offers a practical framework for facilities to engage with the robotics market in a structured, needs-driven manner. The methodology begins by deconstructing operator responsibilities such as temperature checks, abnormal noise detection, leak identification, surface condition monitoring, and analog gauge reading. Each task is analyzed based on its frequency and mode of execution. The study then determines what type of robotic hardware is required: for example, whether a mobile inspection robot with integrated sensors is sufficient or whether a system with a robotic arm or specialized payload is necessary. It also assesses whether such solutions currently exist in the market. Where autonomous execution is possible, the corresponding AI requirement is identified ranging from rule-based logic and image analytics to machine learning for historical trend prediction. For instance, temperature verification is evaluated across multiple approaches: real-time thermal profiling, setpoint-triggered alarms, and anomaly detection using historical data. While previous studies and industry efforts have often taken a technology-first approach or focused narrowly on isolated functions, this study is differentiated by its ground-up, task-driven methodology. It begins with real operator workflows and builds outward to define the robotic hardware, AI capabilities, and market availability needed to meet those tasks. The result is a framework that serves both as a practical guide for oil and gas facilities looking to adopt robotics and as a strategic input for robotics developers aiming to align innovation with the operational realities and unmet needs of the energy sector.

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Introduction

Gas-oil separation plants, crude stabilisation units and other upstream processing facilities are hazardous environments where reliability and safety are paramount. Field operators perform regular rounds to **inspect gauges, listen for abnormal sounds, check for leaks, verify safety equipment and record readings**. These tasks are crucial for preventing incidents but also expose personnel to hazards such as high pressure, flammable gases and extreme temperatures. Over the past decade the oil and gas industry has turned to **mobile robots** to reduce human exposure and improve the frequency and quality of inspections. Legged robots such as ANYmal and Spot, wheeled crawlers like Taurob's Inspector and tracked robots from Rockwell or TotalEnergies can autonomously navigate sites and carry sensor payloads. Research and industrial trials have demonstrated robots reading analog gauges using computer vision [1] [2], measuring temperatures with thermal cameras, detecting gas leaks via infrared and photo-ionisation sensors [3], and using ultrasonic microphones to detect abnormal vibrations . Boston Dynamics' Spot arm prototype has even opened doors and turned valves [5], while automated fire-watch systems such as Blackstar FireSight detect flames and smoke far faster than human fire watchers [4].

Despite these advances, the **adoption of robotics in oil and gas operations remains fragmented and technology-driven**. Most published papers and industrial deployments showcase specific capabilities of a robot reading gauges, finding gas leaks or navigating stairs rather than providing a holistic assessment of how robotic hardware and AI could cover the full spectrum of operator duties. Yet facility managers need a structured roadmap that starts with the human tasks and matches them to technology solutions. Without that, investments may focus on the wrong areas or leave significant value untapped. Our study addresses this gap by **cataloguing every operator task performed in a GOSP and evaluating how each can be automated**. We classify tasks, define the required sensor payloads and AI algorithms, estimate the fraction of workload that could be removed, and highlight where current technology falls short. This paper builds upon earlier work which outlined general opportunities for inspection robots [1] and digital oilfields but did not systematically align operator tasks with robotics solutions.

Methodology

Task Identification

We began by reviewing the **daily, weekly, monthly, quarterly and yearly checklists** used by operators in a large upstream facility. These checklists cover routine surveillance (e.g., walking around and observing equipment), safety compliance (e.g., fire-protection equipment checks), process demand tasks (e.g., sampling and quality control) and corporate programmes (e.g., audits and paperwork). Through interviews with operations staff and a review of standard operating procedures we identified **184 unique tasks**. Examples include:

- **Daily walk-around checks:** verifying abnormal temperature by touch, listening for unusual noises, checking for leaks, reading gauges, checking for fire, and noting abnormal vibration.
- **Monthly safety equipment checks:** inspecting fire hydrants, hose reels and sprinkler systems; checking emergency lights and exit signs; inspecting fire extinguishers; verifying fire blankets and stretchers; and testing alarms.
- **Utility inspections:** checking purge stations, steam traps, utility stations and PIVs (post-indicator valves); reading substation gauges; inspecting generators.

- **Maintenance support:** performing housekeeping, issuing work permits, standby operator duties, fire watch during hot work, lock-out/tag-out (LOTO), scaffolding checks and excavation safety checks.
- **Operational tasks:** chemical sampling, switching over equipment, start-up and shut-down, checking dead legs, unloading chemicals, and turnarounds (T&I).
- **Administrative and training tasks:** paperwork, training on the distributed control system (DCS), safety reviews, innovation suggestions, OIM meetings and controller performance reports.

Each task includes information on **frequency (daily, weekly, monthly, quarterly or yearly)**, the time taken by operators, and the **load percentage** (the fraction of the operators' total workload devoted to that task). The load percentages sum to 100 % for the entire set of tasks.

Robotic Mapping

For each task we asked three questions:

1. **Can a robot perform this task?** The answer could be **Yes** (fully automatable), **Partially** (the robot can assist but a human is still required) or **No** (not suitable for robotics).
2. **What hardware does the robot require?** We considered five core sensor categories: a **temperature camera** (infrared/thermal imager), a **normal camera** (visible light camera), a **sound sensor** (microphone), a **vibration sensor** (accelerometer or contact probe), and a **gas detector** (for leak detection). We also noted when a **manipulator arm** would be needed to operate valves, push buttons or connect hoses.
3. **Is AI required, and if so, which type?** We mapped tasks to AI techniques such as **image analytics** (computer vision for gauge reading and object detection), **thermal analytics** (infrared pattern analysis), **acoustic anomaly detection**, **vibration analysis**, **gas leak detection algorithms**, **autonomous task sequencing** (for start-ups and shut-downs), and more general **anomaly detection** models. For tasks not requiring AI (because a simple threshold alarm suffices), the column is marked **No**.

The mapping also specified the **automation percentage** (100 % for fully automatable tasks, 50 % or 30 % for partially automatable tasks, and 0 % for non-automatable tasks). For partially automatable tasks, we estimated what portion of the operator's time could be saved by robotics. Finally, we recorded **comments**, **market gap observations** and **additional notes** to capture insights and research needs.

Load Removal Calculation

To quantify the impact of robotics on operator workload, we calculated the **load removed by robotics** for each task as follows:

$$\text{Load removed} = \text{Operator load} \times \text{Automation percentage}.$$

Here the operator load is the percentage of total operator time devoted to that task (column 11 in our data), and the automation percentage is 1.0 for fully automatable tasks, 0.5 for partially automatable tasks (50 %), 0.3 for tasks with 30 % automation, and 0 for non-automatable tasks. Summing over all tasks gives the total operator load and the total load removed. We also summarised these figures by frequency and automation category.

Results

Automation Potential Across Tasks

Out of the **184 tasks** catalogued, our analysis found:

Fully automatable: Approximately 113 tasks (≈72 %) can be carried out entirely by robots using cameras, thermal sensors, gas detectors or manipulators, without human intervention.

Partially automatable: 12 tasks (≈8 %) involve mixed activities (e.g., valve operations coupled with manual sampling); robots can handle the routine steps but still require a human to complete part of the job.

Not automatable: 31 tasks (≈20 %) remain beyond the reach of current robots. These include high-judgement activities (fire watch response, permit issuance), dexterous tasks like physical sampling, and certain safety checks that must be performed by trained personnel. Fully automatable tasks typically involve **routine inspection** for example:

- **Reading gauges and indicators:** Robots equipped with normal cameras can capture images of pressure gauges, level indicators and dial positions. Computer-vision algorithms have successfully read fire-extinguisher pressure gauges autonomously [2] and can interpret analog dials on valves and pumps.
- **Visual leak detection:** Thermal cameras identify hot or cold spots indicating gas or fluid leaks, while gas detectors sense hydrocarbons [3]. Robots such as ANYmal X and Taurob can carry both sensors [3].
- **Fire monitoring:** Fire-watch robots can detect flames and smoke much faster than human observers using thermal cameras and AI [4].
- **Substation inspections:** In power utilities, robots with IR cameras have already replaced human inspectors [6]; this capability transfers readily to GOSP utility stations.
- **Monthly equipment checks:** Tasks like checking hose reels, hydrants, sprinkler systems, PIVs, wind socks and emergency lights require only visual confirmation, which robots can perform reliably.

Partially automatable tasks often require some human judgement or physical manipulation. Examples include:

- **Start-up and shut-down sequences:** Robots can operate controls and follow predefined sequences but still require human supervision due to safety and unforeseen conditions. Boston Dynamics' Spot arm has demonstrated valve turning [5], but complex procedures remain challenging.
- **Housekeeping and cleaning:** Autonomous floor scrubbers can clean surfaces, yet robots cannot tidy clutter or organise equipment, so only about half of the housekeeping workload is automatable.
- **Fire watch and standby operator duties:** Robots can monitor for flames, gas leaks and abnormal conditions, but a human must remain on standby to make decisions and intervene. Fire-watch robots such as FireSight still rely on human fire fighters to extinguish the blaze [4].
- **Chemical unloading and dead-leg flushing:** Robots could theoretically connect hoses or open drain valves but require advanced manipulators and safety measures; thus only about 30–50 % automation is realistic.

Non-automatable tasks include administrative and cognitive activities such as issuing work permits, updating paper binders, innovation suggestion meetings, OIM safety reviews and controller performance reports. These tasks rely on human judgement, compliance with regulatory processes, or creative thinking. Training operators on distributed control systems also remains a human-centred activity; though simulations and virtual reality may complement training, robots themselves are not appropriate.

Sensor and AI Requirements

The mapping revealed clear patterns regarding hardware. **Normal cameras** and **thermal cameras** are the workhorses for inspection. They support gauge reading, surface inspection, leak detection and fire monitoring. **Gas detectors** (infrared or electro-chemical sensors) are essential for leak surveys, purge station checks and dead-leg draining. **Sound sensors** help detect abnormal equipment noises and test alarms, while **vibration sensors** are used for equipment condition monitoring. **Manipulator arms** are only required for 11 tasks, mainly those involving physical operation of valves, switches or hoses.

On the AI side, **image analytics** (computer vision) is needed for the majority of fully automatable tasks reading gauges, detecting missing equipment or verifying indicator lights. **Thermal analytics** interpret heat profiles to flag hot bearings or blocked steam traps. **Acoustic analytics** use machine learning to classify normal versus abnormal sounds. **Gas analytics** detect leaks by analysing concentration spikes and, in some cases, estimate leak rates. **Autonomous task sequencing** orchestrates multi-step procedures like start-ups and switchover, ensuring actions are performed in the correct order. **Anomaly detection** models track historical data to predict when equipment may fail or when readings deviate from normal, reducing false alarms and enabling predictive maintenance.

Load Removal Analysis

Summing the operator loads reveals that the 184 tasks together account for **100 %** of the operator workload. Fully automatable tasks contribute **51.1 %** of this load, partially automatable tasks account for **40.2 %**, and non-automatable tasks represent **8.8 %**. Applying the automation percentages yields a **total load removed of 68.9 %**, leaving about one-third of the workload for human operators. Figure 1 summarises the load removed by automation category.

Category	Total Load (%)	Load Removed (%)
Fully automatable	51.08	51.08
Partially automatable	40.17	17.82
Not automatable	8.75	0.00
Total	**100.0**	**68.90**

Breaking the results down by frequency (Table 1) shows that **daily tasks** dominate operator workload and automation opportunities. Daily tasks contribute **86.6 %** of the load, of which robots could remove **61.1 %**. Monthly tasks account for only **5.2 %** of the load but are mostly automatable; weekly tasks account for **5.8 %** of the load with moderate automation potential; quarterly tasks contribute **2.4 %** of the load, while yearly tasks are negligible.

Frequency	Total Load (%)	Removed Load (%)	Fully	Partial	None
Daily	86.56	61.15	20	8	3
Weekly	5.75	2.19	8	4	4
Monthly	5.24	4.48	15	7	2
Quarterly	2.37	1.04	1	2	1
Yearly	0.08	0.04	0	2	1

These results highlight that investment in robotics for **daily inspection tasks** can yield the greatest reduction in operator workload, whereas administrative and low-frequency tasks contribute little to the overall load and are generally not automatable.

Examples of Task Mappings

To illustrate the task-first mapping, several representative tasks are summarised below:

1. **Monthly Sprinkler System Check** Operators confirm that sprinkler valves are open and pressure gauges are within range. Robots can fully automate this by using a normal camera to read gauges and inspect valve indicators. A manipulator arm may press the test valve to ensure water flows. AI uses image analytics to interpret gauge readings and detect misalignment. The automation percentage is 100 %, and the robotic solution removes the full **0.02 %** operator load for this task. There is no dedicated sprinkler inspection robot on the market, so integration of cameras and manipulators into an inspection robot is required.
2. **Monthly Fire Extinguisher Check** This task involves ensuring each extinguisher is present, in the correct location, and fully charged. A robot can read the pressure gauge using computer vision [2] and verify that the safety pin is intact; however, it cannot measure the extinguisher's weight or perform refilling. The task is rated **50 % automatable**; using a camera and AI the robot saves **0.12 %** of operator load but still leaves human tasks such as replacement.
3. **Monthly Steam Trap Check** Operators listen for steam leaks and feel temperature differences across the trap. Robots equipped with a thermal camera and an ultrasonic microphone can detect a stuck trap or a leaking valve. AI compares upstream and downstream temperatures and analyses ultrasonic signatures to detect abnormal behaviour. Because robots cannot physically repair or replace the trap, the task is fully automatable from a monitoring perspective, removing **0.08 %** of the operator load.
4. **Start-up and Shut-down of Equipment).** Starting a pump or shutting down a compressor involves multiple steps, safety interlocks and possible adjustments. A robot with a manipulator can operate buttons and levers and follow a predefined sequence, but human oversight is necessary to handle unexpected conditions. We therefore classify the task as **50 % automatable**, saving **1.89 %** of the load.
5. **Vehicle Inspection** Operators inspect company vehicles daily for safety: checking lights, tyres and general condition. A robot can walk around vehicles, take photographs and use computer vision to detect missing hubcaps, damaged lights or under-inflated tyres. Thermal imaging identifies overheated brakes, while gas sensors detect exhaust leaks. Only **50 %** of the load is removed because some checks (checking oil or coolant levels) require human intervention, but the robot can still save **0.20 %** of workload.
6. **Fire Watch** During hot work, operators must continuously watch for sparks and ignition. Robots such as FireSight can detect flames and smoke in milliseconds [4], using thermal and optical cameras and AI. Robots cannot extinguish a fire or decide when to intervene, so we classify the task as **0 % automatable** (0 % load removed) and recommend combining robotic detection with fixed suppression systems.
7. **Lock-Out/Tag-Out (LOTO) (Task 56).** This safety procedure mandates that a person physically applies locks and tags to energy isolation devices and verifies zero energy before maintenance. Because it is a compliance and accountability process requiring human judgement, **robotic automation is not applicable**. The task's **1.26 %** load must remain with human operators.
8. **Innovation & HSE Suggestions, Near Misses & Stop Work** Encouraging staff to report near misses and suggest improvements is a cultural practice. Robots cannot generate or evaluate suggestions, and automation is not meaningful. We assign 0 % automation and note that digital reporting tools can complement human creativity but not replace it.

Through these examples the advantages and limitations of robots become apparent. Where tasks are repetitive, sensory and objective reading a gauge or scanning for leaks robots excel. Where tasks involve subjective judgement, high dexterity or cognitive processes, human involvement remains essential.

Discussion

Comparison with Existing Literature

Our findings align with, but also extend, previous research on robotics in the oil and gas sector. Chen et al. (2014) surveyed robotic applications on offshore platforms and identified repetitive inspection tasks as prime candidates for automation [1]. They described prototypes of mobile robots reading gauges, opening valves and listening for abnormal sounds, similar to tasks automated in our study. More recent deployments, such as TotalEnergies' ARGOS robot and ANYbotics' ANYmal, have demonstrated autonomous gauge reading, thermal imaging and gas leak detection [2] [3]. Blackstar FireSight's fire-watch robot surpasses human detection times [4], and power utilities use robots for substation patrols [6]. However, these efforts often focus on a **single use case** and rarely quantify the impact on operator workload.

Our operator-centric approach reveals the **breadth of tasks** suited for robotics and highlights those that remain beyond current capabilities. For example, **chemical unloading** tasks require sophisticated manipulation and risk management; only a few experimental systems exist. **Training, administration and cultural activities** (e.g., suggestion programmes) are outside the scope of robotics. By cataloguing these tasks and identifying market gaps, our study provides a more nuanced roadmap than technology-driven demonstrations. It also underscores the need for **integrated fleet management**, **multi-robot collaboration** and **charging infrastructure**. Most existing robots operate individually; coordinating a fleet of 20 or more robots across a large facility will require new software for mission planning, task allocation and charging management. Our study points to this gap as a key research and development priority.

Implications for Facilities and Vendors

For facility managers, the study offers a **practical guide** to prioritise automation investments. Focusing first on daily inspection tasks robots can achieve full automation will yield the greatest reduction in operator workload and improve safety. Deploying robots for monthly and weekly equipment checks can free operators for higher-value work. However, managers should recognise that tasks involving manual manipulation or human judgement will likely remain shared duties for the foreseeable future. Careful change management and training are needed to integrate robots into the operator workforce.

For robotics vendors, the results signal where **future development is most needed**. Market gaps include:

- **Combined sensory payloads**: Many inspection robots carry a standard camera and thermal imager, but few integrate gas detectors, vibration probes and ultrasonic microphones. A modular payload system would enable more tasks to be covered with one platform.
- **Manipulation capability**: Robots capable of opening valves, connecting hoses, or turning large hand wheels remain experimental [5]. Developing robust, ATEX-certified arms and end-effectors is essential for partial automation tasks.
- **Integrated fleet management**: Managing dozens of robots requires scheduling, mission assignment, battery charging and avoidance of conflicts. Current vendors offer basic dashboards but not the sophisticated multi-robot collaboration that plants need.
- **Validation between robots and drones**: In an emergency a drone may detect a leak or fire. Coordinating with a ground robot carrying a gas sniffer to confirm and characterise the leak is not yet supported.
- **Administrative and training solutions**: While robots cannot perform paperwork or training, there is an opportunity for AI-driven software to digitise and automate these processes. Integrating

robotic inspection data into management systems can also support digital twins and predictive maintenance.

Limitations and Future Research

This study has limitations. First, it focuses on a single facility's task list; other plants may have different procedures or additional tasks. Second, the automation percentages are estimated based on current technology and may change as robotics advances. Third, the analysis assumes that robots can operate continuously and reliably; in practice, downtime due to maintenance, environmental conditions or certification issues (e.g., ATEX) could reduce the effective load removed. Finally, the study does not address the **economic analysis** of costs versus benefits; while robotics can reduce operator workload, cost-benefit calculations are necessary for investment decisions.

Future research should explore **multi-robot coordination algorithms**, **adaptive scheduling** and **decision support** to manage fleets in dynamic industrial settings. Advanced AI techniques such as **reinforcement learning** could allow robots to adapt to unstructured environments, while **digital twins** could simulate robotic operations to optimise deployment. Human-robot interaction studies are needed to ensure trust and safety when operators and robots work side by side. Furthermore, case studies of robotic implementation in live plants would provide valuable data on reliability, maintenance and worker acceptance.

Conclusion

This paper presents a comprehensive, operator-centric framework for evaluating how robotics and AI can automate routine tasks in oil and gas facilities. By cataloguing 78 tasks from daily walk-arounds to yearly drills and mapping each to specific sensors, AI techniques and manipulation requirements, we show that more than half of operator tasks can be fully automated, a quarter can be partially automated, and less than one fifth currently require human execution. Robots equipped with cameras, thermal imagers, microphones, gas detectors and manipulators can handle the bulk of routine inspections, leak surveys and equipment checks [2] [3]. Using AI for image, thermal, acoustic and gas analytics allows robots to interpret sensor data, detect anomalies and follow predefined sequences. Our load analysis indicates that up to **68.9 % of operator workload** could be removed by robotics, with the greatest impact coming from daily tasks.

The study also identifies market gaps and research directions. Vendors should develop modular, multi-sensor payloads, robust manipulators and fleet-management software to cover a wider range of tasks. Facilities need to invest strategically, starting with fully automatable tasks and planning for human–robot collaboration where partial automation is possible. Finally, the industry must integrate robotic data into digital twins, condition-monitoring systems and enterprise workflows to realise the full benefits of automation. By starting from the operator's tasks rather than the robot's capabilities, this study lays the groundwork for a future where oil and gas operations are safer, more efficient and increasingly autonomous.

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